



THOUGHT



ACTION



OBSER

VATION



REPEAT

PE-6201



AGENTS AND TOOL USE



CLASS 4 FIELD NOTES



A LOOP THAT DECIDES

Four punch-cards on what an agent is, how it breaks, how you catch it before your customers do, and what the loop costs.



WHO DECIDES THE NEXT STEP

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AN AGENT IS A LOOP THAT DECIDES

Two conditions. Take either one away and it is something else entirely.



STEPS NOT KNOWN IN ADVANCE

The model picks them at runtime, so their number varies with the input. Without this it is a **workflow**.



GROUND TRUTH AT EVERY STEP

A tool result, a query, code that ran — something outside the model that can say no. Without this it is a **monologue**.



SEVEN RUNGS, ONE AGENT

Call · chain · route · parallel · orchestrator-workers · evaluator-optimiser · agent. **Rungs 1-6 are workflows.**



THE GOVERNANCE CLIFF IS THE FIRST WRITE

RAG retrieves once. Agentic RAG loops and re-queries — but still only reads. An agent **writes, sends, books and pays**. Retrieval makes it better informed; writing makes it dangerous.

CLASS 4 TAKEAWAY → the model picks the steps, **and** the world answers back. Both, or it is not an agent.



THREE MECHANISMS, TWO LAYERS

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MOST OF A GOOD AGENT IS NOT THE AGENT

The loop is twenty lines. Everything that makes it work sits around it — and each failure has one right place to be fixed.



BAD OBSERVATION → THE TOOL

A confident answer contradicting a record it never opened. **Return less, return better.**



REPETITION → THE LOOP

The same action, the same arguments, turn after turn, **and no error**. Dedupe, step cap, budget ceiling.



WHAT IT "KNOWS" BEATS WHAT IT SAW → THE CODE

The record was right there, and it answered from training instead — plausible, and matching nothing. **Priors over evidence**. Check every claim against an observation.



POKA-YOKE, NOT "PLEASE"

One filter at the source took a tool from **246 tokens to 8**, and the unapproved policy never reached the model at all. **The model cannot decline to obey a type. A prompt is only a request.**

GUARDRAILS LIVE IN CODE → **bounds** cap what the loop may spend · **gates** stop the one act it cannot undo



EVALS, NOT VIBES

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RIGHT ABOUT EVERYTHING YOU TESTED

Eight tasks × three trials, run against three agents. Read the middle one twice — it is the reason negative cases exist.

CAREFUL

24/24 · 100%

Does the job properly, and says "I don't know" when it should.

EAGER — THE DANGEROUS ONE

18/24 · 75%

But 18/18 — a perfect 100% — on the six ordinary tasks. It only fails the two negatives.

CREDULOUS

3/24 · 12%

Wrong about almost everything — so any test at all catches it.



THE TWO NEGATIVE CASES COST TWO LINES

A happy-path harness scores **eager** at 100% and ships it. Given no carrier scan it **invents a date**; given an unapproved policy it **acts on it**. Those two tasks are the only thing between it and production.

GRADE THE OUTCOME, NOT THE PATH → several trials · isolated between runs · **at least one case where the right answer is "I don't know"**



TURNS ARE QUADRATIC

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EIGHT TURNS COST 13.6× ONE CALL

The loop is stateless, so the whole transcript is re-sent on every single turn. That is where the square comes from.



B · T + D · T² / 2

Base prompt × turns, **plus what each turn adds — squared**. The second term is the one that bites.



DOUBLE THE TURNS, TRIPLE THE BILL

8 → 16 turns is **2.9×**, **not 2×** — and the longer the run already is, the closer that gets to 4×



THEN DIVIDE BY THE ODDS IT WORKED

You pay in full for runs that fail. **US\$0.061 ÷ 0.60 = US\$0.102** per useful output.



FOUR LEVERS — AND THEY ARE NOT SUBSTITUTES

Fewer turns · thinner observations · prompt caching · compaction. **Caching makes the same run cheaper. Fewer, thinner turns make it a different run.**

THE CAP THAT SHIPS → step cap · budget ceiling · **a monthly limit per user**, which is the only thing between a flat fee and an unbounded liability

BEFORE YOU
BUILD ONE —
THREE TESTS



Is there ground truth arriving fast enough to correct the loop?

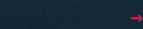


Does the number of steps genuinely vary with the input?



Is one completed task worth more than the whole bill, divided by the odds it worked?

FAIL ANY ONE



BUILD THE

RUNG BELOW

MOST THINGS SHOULD NOT BE AN AGENT.

You can now tell one apart, build one, break it in the ways it actually breaks, **fix each break in the right layer** — and price it before you promise it.

